Deep Multimanifold Transformation-Based Multivariate Time Series Fault Detection

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***Abstract*—Unsupervised fault detection in multivariate time series (MTS) plays a vital role in ensuring the stable operation of complex systems. Traditional methods often assume that normal data follow a single Gaussian distribution and identify anomalies as deviations from this distribution. However, this simplified assumption fails to capture the diversity and structural complexity of real-world time series, which can lead to misjudgments and reduced detection performance in practical applications. To address this issue, we propose a new method that combines a neighborhood-driven data augmentation strategy with a multimanifold representation learning framework. By incorporating information from local neighborhoods, the augmentation module can simulate contextual variations of normal data, enhancing the model's adaptability to distributional changes. In addition, we design a structure-aware feature learning approach that encourages natural clustering of similar patterns in the feature space while maintaining sufficient distinction between different operational states. Extensive experiments on several public benchmark datasets demonstrate that our method achieves superior performance in terms of both accuracy and robustness, showing strong potential for generalization and real-world deployment.**

***Index Terms*—Data augmentation, fault detection, multivariate time series, unsupervised soft contrastive learning (CL).**

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I. Introduction

UN SUPERVISED fault detection [15], [23] in multivariate time series (MTS) has become increasingly vital in both academic research and practical domains, especially within industrial environments and the management of large-scale systems [21], [29], [36]. By eliminating the need for labeled data, it enables early identification of anomalies and incipient faults [6], [38], which is crucial for minimizing maintenance overhead and averting potential system failures [17]. This capability makes it an indispensable tool for real-time monitoring and predictive maintenance in scenarios where labeled samples are scarce or entirely unavailable [40].

To address this challenge, a wide range of approaches have been explored, from classical statistical methods to modern deep learning-based models [32], [37]. Despite the notable progress made in recent years [20], most existing techniques still exhibit unsatisfactory detection performance in practical applications, characterized by frequent false positives and missed anomalies (see Tables I and II). This is largely due to the restrictive assumptions imposed during modeling, often necessitated by the absence of supervision. A prevailing strategy is to model normal behavior as a Gaussian distribution and to flag deviations as anomalies [39], as illustrated in Fig. 1(a). While computationally convenient, such assumptions fail to capture the intrinsic diversity and nuanced dynamics of real-world systems.

In reality, both normal and anomalous behaviors often comprise multiple heterogeneous subpatterns, each exhibiting distinct temporal and structural traits [Fig. 1(b)]. Relying on a single Gaussian model obscures these internal variations, leading to suboptimal representations and a reduced capacity to detect subtle deviations [23]. This oversimplification not only impairs the model's ability to capture gradual state transitions but also undermines robustness in highly variable environments [10], [25]. Consequently, there is a pressing need to move beyond Gaussian-centric assumptions and adopt more expressive modeling paradigms capable of characterizing complex, multimodal system behaviors.

To address the limitations caused by the oversimplified Gaussian distribution assumption, we propose an unsupervised framework, termed deep multimanifold transformation for fault detection (DMTFD). Unlike prior methods that characterize normal states with a single Gaussian distribution [39], DMTFD adopts a multimanifold assumption, which

TABLE I

FAULT DETECTION PERFORMANCE OF AUROC ON FIVE PUBLIC DATASETS. WE COMPARE THE PERFORMANCE OF DIFFERENT METHODS USING A CONSISTENT WINDOW SIZE (DMTFD) WITH OUR OPTIMAL RESULTS OBTAINED THROUGH VARYING WINDOW SIZES (DMTFD*) TO SHOWCASE THE MAXIMUM POTENTIAL. THE BEST RESULTS ARE GIVEN IN BOLD

Methods	Journal-Year	Area Under the Receiver Operating Characteristic Curve (AUROC) SWaT	WADI	PSM	MSL	SMD	SMAP	Average	Rank
DROCC [12]	ICML, 2020	72.6(±3.8)	75.6(±1.6)	74.3(±2.0)	53.4(±1.6)	76.7(±8.7)	70.1(±3.5)	70.4	9
DeepSAD [26]	ICLR, 2020	75.4(±2.4)	85.4(±2.7)	73.2(±3.3)	61.6(±0.6)	85.9(±11.1)	72.8(±4.1)	77.4	8
USAD [4]	KDD, 2020	78.8(±1.0)	86.1(±0.9)	78.0(±0.2)	57.0(±0.1)	86.9(±11.7)	74.5(±2.9)	76.9	7
GANF [9]	ICLR, 2022	79.8(±0.7)	90.3(±1.0)	81.8(±1.5)	64.5(±1.9)	89.2(±7.8)	77.6(±3.2)	80.5	6
Autoformer [30]	NeurIPS, 2022	81.2(±1.1)	89.5(±1.3)	83.7(±1.8)	66.0(±1.5)	89.8(±7.1)	78.3(±3.1)	81.4	5
MTGFlow [39]	AAAI, 2023	84.8(±1.5)	91.9(±1.1)	85.7(±1.5)	67.2(±1.7)	91.3(±7.6)	80.2(±2.8)	83.5	3
MTGFlow_C [40]	TKDE, 2024	83.1(±1.3)	91.8(±0.4)	87.1(±2.4)	68.2(±2.6)	91.6(±7.3)	81.1(±2.7)	84.8	2
USD [23]	ICASSP, 2025	82.7(±1.0)	90.5(±1.2)	84.5(±1.3)	66.8(±2.0)	90.5(±6.8)	79.5(±3.2)	82.4	4
DMTFD	OURS	90.5(±0.9)	94.3(±0.4)	**89.2(±2.0)**	75.0(±2.2)	**95.0(±6.1)**	85.7(±2.5)	88.3	1
DMTFD*	OURS	**92.5(±1.2)**	**95.2(±1.1)**	**89.2(±2.1)**	**76.5(±1.0)**	**95.0(±6.4)**	**87.4(±2.1)**	**89.3**	1*

TABLE II

ANOMALY DETECTION PERFORMANCE OF PR ON FIVE PUBLIC DATASETS. WE COMPARE THE PERFORMANCE OF DIFFERENT METHODS USING A CONSISTENT WINDOW SIZE (DMTFD) WITH OUR OPTIMAL RESULTS OBTAINED THROUGH VARYING WINDOW SIZES (DMTFD*) TO SHOWCASE THE MAXIMUM POTENTIAL. THE BEST RESULTS ARE GIVEN IN BOLD

Methods	Journal-Year	Area Under the Precision-Recall Curve (AUPRC) SWaT	WADI	PSM	MSL	SMD	SMAP	Average	Rank
DROCC [12]	ICML, 2020	26.4 (±9.8)	16.7 (±7.1)	60.7 (±11.4)	13.2 (±0.9)	20.7 (±15.6)	31.2 (±5.3)	28.2	9
DeepSAD [26]	ICLR, 2020	45.7 (±12.3)	23.1 (±6.3)	66.7 (±10.8)	26.3 (±1.7)	61.8 (±21.4)	45.3 (±8.4)	44.8	7
USAD [4]	KDD, 2020	18.8 (±0.6)	19.8 (±0.5)	57.9 (±3.6)	31.3 (±0.0)	68.4 (±19.9)	42.5 (±9.5)	39.8	8
GANF [9]	ICLR, 2022	21.6 (±1.8)	39.0 (±3.1)	73.8 (±4.7)	31.1 (±0.2)	64.4 (±21.4)	50.7 (±4.3)	46.8	6
Autoformer [30]	NeurIPS, 2022	35.1 (±7.2)	39.3 (±3.9)	71.4 (±6.0)	29.5 (±2.5)	60.2 (±18.1)	48.2 (±8.6)	47.3	5
MTGFlow [39]	AAAI, 2023	38.6 (±6.1)	42.2 (±4.9)	76.2 (±4.8)	31.1 (±2.6)	64.2 (±21.5)	52.6 (±6.3)	50.8	3
MTGFlow_C [40]	TKDE, 2024	41.2 (±4.8)	42.4 (±5.2)	76.1 (±4.9)	31.2 (±2.8)	64.5 (±20.1)	53.1 (±8.9)	51.4	2
USD [23]	ICASSP, 2025	37.4 (±5.9)	40.5 (±4.1)	74.2 (±5.6)	30.7 (±2.4)	63.3 (±18.4)	51.8 (±6.1)	49.6	4
DMTFD	OURS	53.6 (±10.1)	45.7 (±4.1)	83.2 (±4.9)	32.1 (±3.5)	73.8 (±16.7)	60.2 (±7.2)	58.1	1
DMTFD*	OURS	**59.3** (±6.5)	**60.5** (±3.2)	**83.2** (±3.9)	**46.2** (±1.8)	**73.8** (±13.7)	**65.3** (±8.9)	**64.7**	**1***

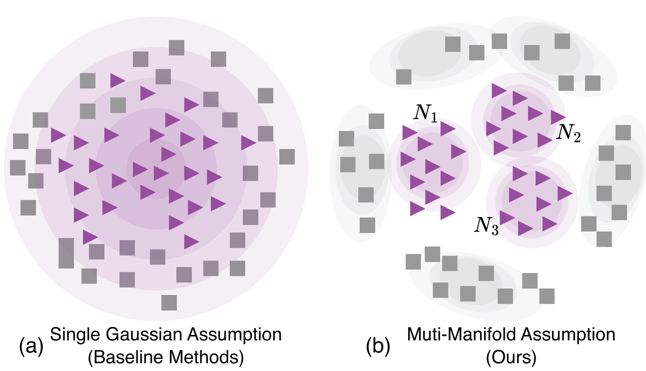

Fig. 1. Motivation of DMTFD. (a) Existing methods rely on Gaussian assumptions and fail to capture complex latent structures with multiple substates. (b) Our proposed DMTFD introduces a multimanifold assumption to better model state variations and improve fault detection accuracy.

ones. To construct a meaningful representation space under this assumption, we introduce a neighbor-aware augmentation strategy that generates context-consistent variants of each sample [18] and adopt a softened similarity constraint to gently pull semantically close instances. This design alleviates the rigidity of binary contrastive signals and supports more nuanced anomaly boundary modeling [41].

Comprehensive experiments show that DMTFD significantly outperforms existing methods in both AUC and precision-recall (PR) metrics. On benchmark datasets, DMTFD achieves consistently lower false alarm rates and higher detection accuracy.

Furthermore, our visualization analysis supports the plausibility of the multi-Gaussian assumption and illustrates how soft contrastive learning (CL) effectively captures underlying substate structures.

acknowledges the intrinsic diversity of normal operating conditions across different system states [see Fig. 1(b)].

Concretely, DMTFD leverages a multimanifold transformation module that maps MTS into a latent space where heterogeneous local patterns are better disentangled and aligned. This transformation enables the model to preserve fine-grained distinctions between submanifolds of normal behaviors while amplifying their deviation from anomalous

1) *Multimanifold Modeling of Operational States:* We break away from the conventional single-Gaussian assumption and adopt a multimanifold view that reflects the heterogeneity of both normal and abnormal conditions. This insight leads to a more faithful and flexible modeling of system behaviors.
2) *New Framework for Unsupervised Representation:* We propose a novel framework that integrates

neighbor-based data augmentation and loss function to construct a smooth and discriminative latent space.

3) *Superior Empirical Performance Across Benchmarks:* Our method achieves over 5% improvement in detection metrics compared to existing baselines, demonstrating strong robustness and generalization in real-world industrial scenarios.

II. Related Work

A. Time Series Anomaly Detection

In time series anomaly detection, one-class classification (OCC) methods—such as USAD [4] and DAEMON [8]—typically assume access to purely normal data during training [5], [16]. GANF [9] employs graph neural networks (GNNs) for anomaly detection in MTS. MTGFlow [39] combines dynamic graphs with normalizing flows, while MTGFlow-cluster [40] further boosts accuracy by clustering entities. AnomalyLLM [22] leverages large language models for knowledge distillation, using prototypical signals and synthetic anomalies to achieve an improvement on UCR datasets across 15 benchmarks. PeFAD [31] introduces PLM-based parameter-efficient federated learning with anomaly masking and synthetic distillation, improving performance by up to 28.74%. Graph-MoE [15] enhances GNN-based detection via expert mixtures and memory routers, effectively utilizing hierarchical features. In addition, prior work USD [23] proposes a CL framework tailored for MTS fault detection, addressing limitations of hard contrastive loss in the presence of view-level noise. Building upon this, the present work introduces a multimanifold transformation strategy and a generalized similarity modeling approach, enabling more expressive representations and significantly improving detection performance and robustness across benchmarks.

B. Contrastive Learning and SoftCL

Soft CL (SoftCL) [33], [35] and deep manifold learning [19], [24] represent two advanced methodologies in the domain of machine learning, particularly in the tasks of unsupervised learning and representation learning. These techniques aim to leverage the intrinsic structure of the data to learn meaningful and discriminative features. At its core, SoftCL [28], [33], [35] is an extension of the CL framework [7], [13], [14], which aims to learn representations by bringing similar samples closer and pushing dissimilar samples apart in the representation space. However, SoftCL introduces a more nuanced approach by incorporating the degrees of similarity between samples into the learning process, rather than treating similarity as a binary concept. This is achieved through the use of soft labels or continuous similarity scores, allowing the model to learn richer and more flexible representations. The softness in the approach accounts for the varying degrees of relevance or similarity among data points, making it particularly useful in tasks where the relationship between samples is not strictly binary or categorical, such as in semi-supervised learning or in scenarios with noisy labels.

III. Preliminary

Normalizing flow is an unsupervised density estimation approach to map the original distribution to an arbitrary target distribution by a stack of invertible affine transformations. When density estimation on original data distribution $\mathcal{X}$ is intractable, an alternative option is to estimate z density on target distribution $\mathcal{Z}$. Specifically, suppose a source sample $x \in \mathcal{R}^D \sim \mathcal{X}$ and a target distribution sample $z \in \mathcal{R}^D \sim \mathcal{Z}$. Bijective invertible transformation $\mathcal{F}_\theta$ aims to achieve one-to-one mapping $z = f_\theta(x)$ from $\mathcal{X}$ to $\mathcal{Z}$. According to the change of variable formula, we can get

$$P_{\mathcal{X}}(x) = P_{\mathcal{Z}}(z) \left|\det \frac{\partial f_\theta}{\partial x^T}\right|. \tag{1}$$

Benefiting from the invertibility of mapping functions and tractable Jacobian determinants $|\det(\partial f_\theta/\partial x^T)|$, the objective of flow models is to achieve $\hat{z} = z$, where $\hat{z} = f_\theta(x)$.

Flow models are able to achieve more superior density estimation performance when additional conditions C are input [3]. Such a flow model is called conditional normalizing flow, and its corresponding mapping is derived as $z = f_\theta(x|C)$. Parameters θ of f_θ are updated by maximum likelihood estimation (MLE)

$$\theta^* = \arg\max_\theta \left(\log\left(P_{\mathcal{Z}}\left(f_\theta(x|C)\right)\right) + log\left(\left|\det \frac{\partial f_\theta}{\partial x^T}\right|\right)\right). \tag{2}$$

IV. Methodology

A. Notation and Problem Definition

Consider an MTS dataset $\mathbf{X}$. The dataset encompassing K entities, each with L observations, is denoted as $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_K)$, where each $\mathbf{x}_k \in \mathbb{R}^L$. Z-score normalization is employed to standardize the time series data across different entities. A sliding window approach, with a window size of T and a stride size of S, is utilized to sample the normalized MTS, generating training samples $\mathbf{x}^c$, where c denotes the sampling count and $\mathbf{x}^c$ represents the segment $\mathbf{x}_{[cS-T/2]:[cS+T/2]}$.

The objective of unsupervised fault detection is to identify segments $\mathbf{x}^c$ within $\mathbf{X}$ exhibiting anomalous behavior. This process operates under the premise that the normal behavior of $\mathbf{X}$ is known, and any significant deviation from this behavior is considered abnormal. Specifically, abnormal behavior is characterized by its occurrence in low-density regions of the normal behavior distribution, defined by a density threshold $\theta < \rho(\text{normal behavior})$. The task of unsupervised fault detection in MTS can thus be formalized as identifying segments $\mathbf{x}^c$ where $\rho(\mathbf{x}^c) < \theta$.

Definition 1: (Supervised Fault Detection in MTS): Let $\mathcal{D} = \{(\mathbf{x}^c, y^c)\}_{i=1}^N$ be a labeled dataset for an MTS, where each segment $\mathbf{x}^c$ is annotated with a label $y^c \in \{0, 1\}$, indicating normal (0) or abnormal (1) behavior. Supervised fault detection aims to learn a function $f : \mathbb{R}^L \rightarrow \{0, 1\}$ that can accurately classify new, unseen segments $\mathbf{x}_{\text{new}}$ as normal or abnormal based on learned patterns of faults.

Definition 2: (Unsupervised Fault Detection under Gaussian Assumption): In the Gaussian assumption context, unsupervised fault detection in an MTS operates on the premise that

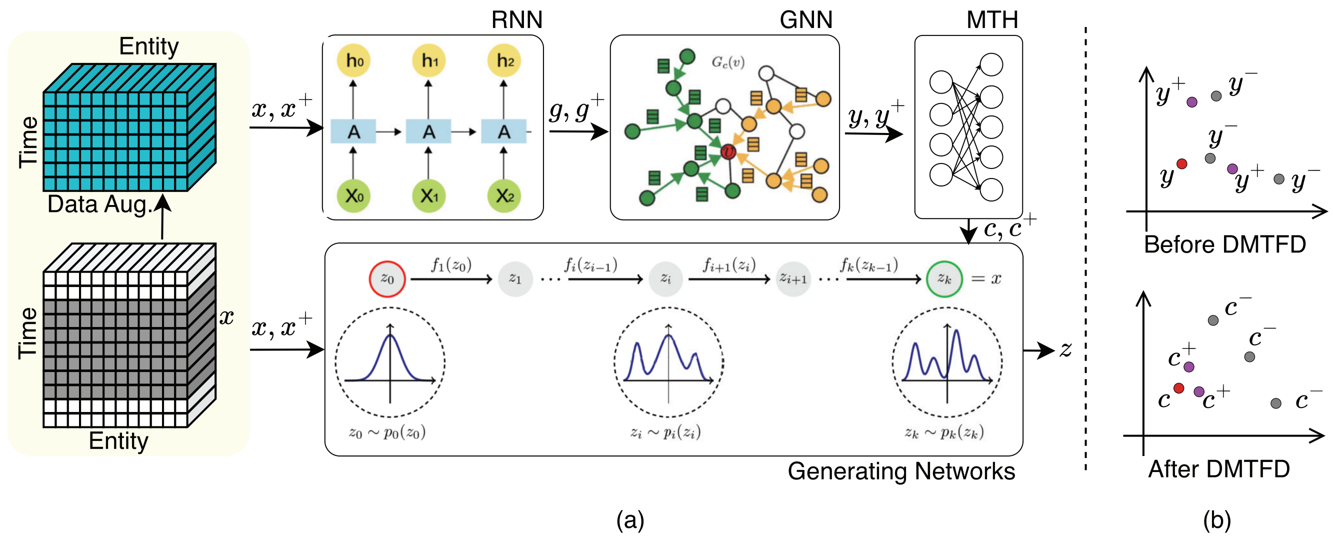

Fig. 2. Overview of the proposed DMTFD framework. (a) Temporal sequences x and x^+ are processed via RNN to capture temporal dynamics, followed by a GNN to model interentity dependencies, yielding embeddings y. A SoftCL head maps y to transformed features c, which are then used for density modeling via a flow-based generative network. (b) SCL head helps cluster similar samples and separate dissimilar ones in the feature space.

the distribution of normal behavior can be modeled using a Gaussian distribution. Anomalies are identified as segments $\mathbf{x}^c$ that fall in regions of low probability under this Gaussian model, specifically where $\rho(\mathbf{x}^c) < \theta$, with θ being a predefined density threshold.

Definition 3: (Unsupervised Fault Detection under Multi-Gaussian Assumption): Under the multi-Gaussian assumption, unsupervised fault detection recognizes that the distribution of normal behavior in an MTS may encompass multiple modes, each fitting a Gaussian model. This assumption allows for a more nuanced detection of anomalies, which are segments $\mathbf{x}^c$ that do not conform to any of the modeled Gaussian distributions of normal behavior, detected through a composite density threshold criterion $\rho(\mathbf{x}^c) < \theta$.

B. Neural Network Structure of DMTFD

The core idea of **DMTFD** is to learn precise and discriminative latent representations by integrating data augmentation with SoftCL. As illustrated in Fig. 2, the DMTFD framework jointly leverages a recurrent neural network (RNN) $R(\cdot;\theta)$, a GNN $G(\cdot;\phi)$, and a normalizing flow model $F(\cdot;\alpha)$ to effectively model the temporal patterns, relational structures, and distributional properties of MTS data for fault detection.

1) RNN-GNN-Flow Architecture: To jointly capture temporal dynamics, interentity dependencies, and density patterns, we adopt an RNN-GNN-Flow architecture. Given the multivariate input at time window c, denoted as x^c, we first compute the temporal encoding $g^c = R(x^c;\theta)$ using an RNN parameterized by θ; a GNN with parameters ϕ then produces the spatiotemporal embedding $y^c = G(g^c;\phi)$; and finally, a normalizing flow with parameters α estimates the log-likelihood score $\ell^c = F(y^c;\alpha)$, which is used to detect anomalies—where low likelihood indicates high abnormality.

2) Manti-Manifold Transformation Head (MTH): To enhance the discriminative quality of the learned representations, we introduce a multimanifold transformation head, implemented as a multilayer perceptron (MLP) $H(\cdot;\omega)$. This module receives the spatiotemporal embedding y^c from the GNN as input and projects it into a latent feature space optimized for CL $z^c = H(y^c;\omega)$, where ω denotes the learnable parameters of the MLP and z^c is the contrastive representation corresponding to time window c. The goal of this module is to enforce semantic alignment between similar instances while ensuring separation among dissimilar ones. This is achieved via a soft contrastive loss, which penalizes embeddings of positive pairs (semantically similar samples) that are far apart and encourages larger distances for negative pairs (semantically dissimilar samples) that are too close. Unlike hard contrastive formulations that impose strict binary similarity, the soft version allows for graded similarity relationships, thereby providing a smoother optimization landscape and improving robustness to intraclass variability.

C. MLE Loss and Data Augmentation

A key objective of the DMTFD framework is to accurately identify abnormal behaviors by modeling the underlying distribution of normal operational data. To this end, we employ *MLE* as the training criterion for the flow-based density estimation module. The core motivation behind this design is to encourage the model to assign high likelihoods to normal patterns while assigning low likelihoods to anomalous deviations, thus enabling precise and fine-grained fault detection.

Formally, given the contrastive representation z^c for time window c, the normalizing flow transformation $f(\cdot;\alpha)$ maps it into a latent variable $\hat{z}^c = f(z^c;\alpha)$ that follows a known target distribution (typically standard Gaussian with mean μ

and identity covariance). The MLE loss is then defined as

$$\mathcal{L}_{\mathrm{MLE}} = \frac{1}{N}\sum_{c=1}^{N}\left[-\frac{1}{2}\|\hat{z}^c-\mu\|_2^2 + \log\left|\det\left(\frac{\partial f_\alpha}{\partial z^c}\right)\right|\right] \tag{3}$$

where N is the number of time windows and μ is the mean of the target Gaussian distribution $\mathcal{Z}$. The first term encourages $\hat{z}^c$ to match the target distribution, while the second term—i.e., the log-determinant of the Jacobian—ensures that the flow transformation remains invertible and probability-preserving. By optimizing $\mathcal{L}_{\mathrm{MLE}}$, the flow model learns to model the density landscape of normal patterns precisely. At inference time, data points with significantly low log likelihoods under the learned distribution are flagged as anomalies, enabling accurate and interpretable fault detection across diverse temporal and relational contexts.

Data augmentation is crucial for improving the model's ability to generalize beyond the training data by artificially expanding the dataset's size and diversity [34]. Our methodology incorporates two primary strategies: neighborhood discovery and linear interpolation.

1) Neighborhood Discovery: This technique focuses on leveraging the local structure of the data to generate new, plausible data points that conform to the existing distribution. For each data point x^c, we define its neighborhood $N(x^c)$ using a distance metric $d(\cdot,\cdot)$, such as the Euclidean distance. The neighborhood consists of points x^c_{new} that meet the criterion

$$d\left(x^c, x^c_{\mathrm{new}}\right) \le \epsilon \tag{4}$$

where ϵ is a predetermined threshold defining the neighborhood's radius. This method captures the local density of the data, enabling the generation of new samples within these densely populated areas and thus enhancing the dataset with variations that align with the original data distribution.

2) Data Augmentation by Linear Interpolation: Data augmentation strategy augments the dataset by creating intermediate samples between existing data points. For two points x^c and x^c_{new}, a new sample x^c_{new} is formulated as

$$x^c_{\mathrm{new}} = \alpha x^c + (1-\alpha)\, x^c_{\mathrm{new}} \tag{5}$$

where α is a random coefficient sampled from a uniform distribution, $\alpha \sim U(0,1)$. This approach facilitates a smooth transition between data points, effectively bridging gaps in the data space and introducing a continuum of sample variations. By interpolating between points either within the same neighborhood or across different neighborhoods, we substantially enhance the dataset's diversity and coverage, providing the model with a more comprehensive set of examples for training.

D. MML With Data Augmentation

In data-augmentation-based CL, the task is framed as a binary classification problem over pairs of samples. Positive pairs, drawn from the joint distribution $(x^{c1}, x^{c2}) \sim P_{x^{c1},x^{c2}}$, are labeled as $\mathcal{H}_{c1,c2} = 1$, whereas negative pairs, drawn from the product of marginals $(x^{c1}, x^{c2}) \sim P_{x^{c1}}P_{x^{c2}}$, are labeled as $\mathcal{H}_{ck} = 0$. The goal of CL is to learn representations that maximize the similarity between positive pairs and minimize it between negative pairs, utilizing the InfoNCE loss

$$\mathcal{L}_{\mathrm{CL}}(x^{\mathrm{c1}}, x^{\mathrm{c2}}, \{x_{\mathrm{cn}}\}_{\mathrm{cn}=1}^{N_K}) = -\log\frac{\exp\left(z^{\mathrm{c1}\,T}z^{\mathrm{c2}}\right)}{\sum_{k=1}^{N_K}\exp\left(z^{\mathrm{c1}\,T}z_{\mathrm{cn}}\right)} = -\log\frac{\exp\left(S\left(z^{\mathrm{c1}},z^{\mathrm{c2}}\right)\right)}{\sum_{k=1}^{N_K}\exp\left(S\left(z^{\mathrm{c1}},z_{\mathrm{cn}}\right)\right)} \tag{6}$$

where $(x^{\mathrm{c1}}, x^{\mathrm{c2}})$ constitutes a positive pair and $(x^{\mathrm{c1}}, x^{\mathrm{cn}})$ a negative pair, with $z^{\mathrm{c1}}, z^{\mathrm{c2}}$, and z^{cn} being the embeddings of $x^{\mathrm{c1}}, x^{\mathrm{c2}}$, and z^{cn}, respectively, and N_K representing the number of negative pairs. The similarity function $S(z^{\mathrm{c1}}, z^{\mathrm{c2}})$ is typically defined using cosine similarity. This method effectively enhances the model's discriminative power by distinguishing between closely related (positive) and less related (negative) samples within the augmented data space.

The conventional CL (CCL) loss is structured around a single positive sample contrasted against multiple negatives. To refine this, we have restructured the CCL loss into a more nuanced form that utilizes labels for positive and negative samples, denoted by $\mathcal{H}_{\mathrm{c1,c2}}$. Detailed explanations of this transformation from (6) and (7) are available in Supplementary Material A.

$$\mathcal{L}_{\mathrm{CCL}}(x^{\mathrm{c1}}, \{x^{\mathrm{c2}}\}_{\mathrm{c2}=1}^{N_K}) = -\sum_{j=1}\left\{\mathcal{H}_{\mathrm{c1,c2}}\log Q_{\mathrm{c1,c2}} + \left(1-\mathcal{H}_{\mathrm{c1,c2}}\right)\log \dot{Q}_{\mathrm{c1,c2}}\right\} \tag{7}$$

where $\mathcal{H}_{\mathrm{c1,c2}}$ indicates if samples $c1$ and $c2$ have been augmented from the same source. $\mathcal{H}_{\mathrm{c1,c2}} = 1$ signifies a positive pair $(x^{\mathrm{c1}}, x^{\mathrm{c2}})$, and $\mathcal{H}_{\mathrm{c1,c2}} = 0$ indicates a negative pair. The term $Q_{\mathrm{c1,c2}} = \exp(S(z^{\mathrm{c1}}, z^{\mathrm{c2}}))$ represents the density ratio, as defined and computed by the backbone network. To enhance robustness against view-level noise introduced by data augmentation, we propose the *multimanifold loss (MML)*. Unlike conventional contrastive losses that treat pairwise labels as binary constants, MML introduces soft similarity-based weights, enabling the model to downweight uncertain or noisy samples during training.

Given an anchor sample x^{c_1} and a set of associated samples $\{x^{c_2}\}_{c_2=1}^{N_K}$, the MML loss is defined as

$$\mathcal{L}_{\mathrm{MML}}(x^{c_1}, \{x^{c_2}\}) = -\sum_{c_2=1}^{N_K}\left[P_{c_1,c_2}\log Q_{c_1,c_2} + \left(1-P_{c_1,c_2}\right)\log\left(1-Q_{c_1,c_2}\right)\right] \tag{8}$$

where P_{c_1,c_2} is the soft label (weight) reflecting the similarity in input space and Q_{c_1,c_2} denotes the similarity in the contrastive latent space. They are defined as

$$P_{c_1,c_2} = \begin{cases} e^{\alpha}\cdot\kappa(y^{c_1}, y^{c_2}) & \text{if } \mathcal{H}_{c_1,c_2} = 1 \\ \kappa(y^{c_1}, y^{c_2}) & \text{otherwise} \end{cases}$$
$$Q_{c_1,c_2} = \kappa(z^{c_1}, z^{c_2}) \tag{9}$$

where $\mathcal{H}_{c_1,c_2} \in \{0,1\}$ indicates whether x^{c_2} is a positive pair of x^{c_1}, $\alpha \in [0,1]$ is a confidence prior that emphasizes positive pairs, and $\kappa(\cdot,\cdot)$ is a similarity kernel.

To improve robustness against augmentation noise and sample-level variability, we adopt a generalized Gaussian

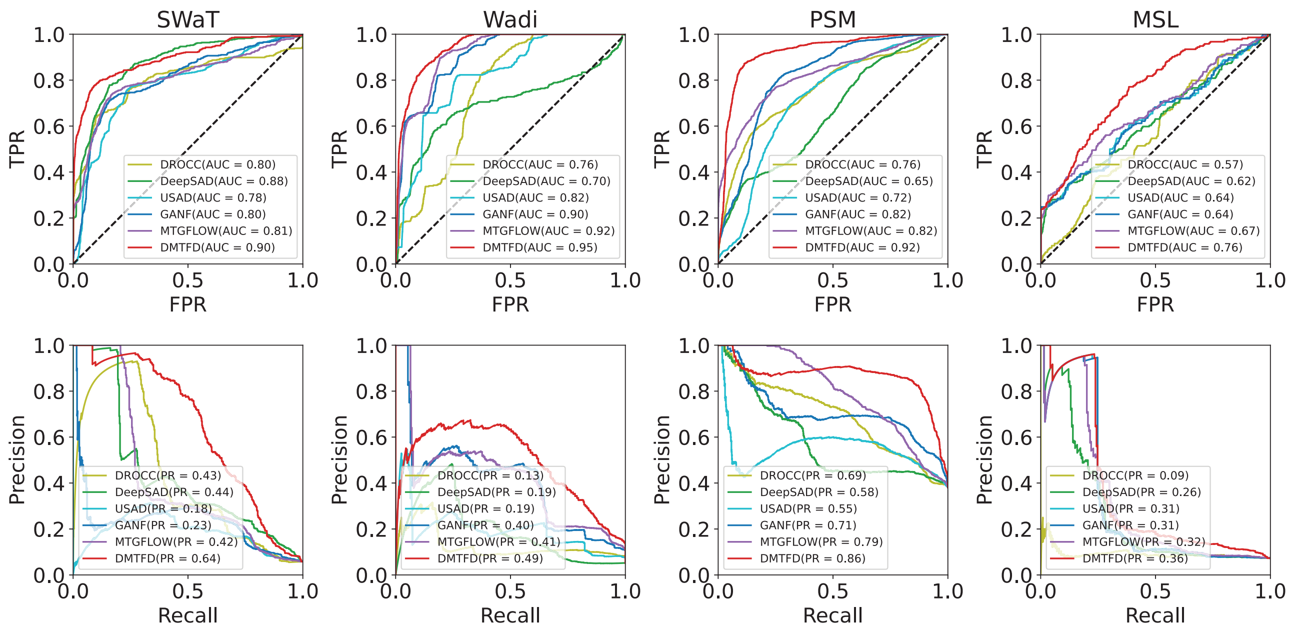

Fig. 3. ROC and PR plots of four datasets. The ROC and PR curves of the SWaT, WADI, PSM, and MSL datasets are shown. The ROC curves illustrate the tradeoff between true positive rate and false positive rate, while the PR curves show the tradeoff between precision and recall. The performance of our method is compared against SOTA methods, demonstrating the effectiveness of our approach.

kernel $\kappa^{\beta}(\cdot,\cdot)$ as the similarity function in MML. Compared to the standard Gaussian kernel, the generalized Gaussian provides a tunable shape parameter that allows for heavier tails when needed, enabling the model to assign meaningful similarity scores even to moderately distant pairs. The kernel is defined as

$$\kappa^{\beta}(a,b) = \exp\left(-\left(\frac{\|a-b\|_2}{\sigma}\right)^{\beta}\right) \tag{10}$$

where $\beta > 0$ controls the shape of the decay (with $\beta = 2$ reducing to a Gaussian kernel and $\beta < 2$ producing heavier tails) and σ is a scale parameter. This formulation makes the loss function more tolerant to view-level noise and nonuniform sample structures, leading to smoother gradients and improved generalization under multimanifold distributions. Unlike USD [23], which relies on a fixed Gaussian assumption and sharp contrastive margins, our generalized kernel formulation offers adaptive flexibility across manifold structures and improves tolerance to cross-view variation. Detailed comparisons and analyses are provided in Supplementary Material A.

E. Optimization Objectives

The entire DMTFD framework, encompassing the RNN, GNN, and FLOW models, is optimized jointly through MLE to ensure effective anomaly detection. The joint optimization process is formulated as follows:

$$L_{\mathrm{DMTFD}} = \mathcal{L}_{\mathrm{MLE}} + \mathcal{L}_{\mathrm{MML}} \tag{11}$$

where $\mathcal{L}_{\mathrm{MLE}}$ is the MLE loss and $\mathcal{L}_{\mathrm{MML}}$ is the MML loss. The MLE loss is designed to maximize the likelihood of observing the transformed data points $\hat{z}_i$ under the model, while the MML loss is designed to enhance the model's sensitivity to subtle differences between normal and abnormal patterns. By embedding these mathematical formulations into each step of the DMTFD framework, we ensure a rigorous approach to modeling and detecting anomalies in MTS data, setting a new standard for unsupervised fault detection in complex systems.

V. Experiments

A. Dataset Information and Implementation Details

Datasets Information: We evaluate DMTFD on six widely used MTS fault detection datasets (Table A-1 in the Supplementary Material). **SWaT** [11] and **WADI** [2] are collected from water treatment and distribution testbeds, simulating industrial cyberattacks with labeled anomalies. **PSM** [1] contains server metrics from eBay, used to detect system performance anomalies. **MSL** [16] provides telemetry from NASA's Curiosity rover for space mission anomaly detection. **SMD** [27] is from a large-scale server farm and captures system behavior across multiple machines; we report average results and ensure that test sets include anomalies via fixed random seeds. SMAP is a **NASA** dataset [16] that contains data from a satellite's attitude control system, used for detecting anomalies in space missions. All datasets are standard in OCC for time series anomaly detection.

1) Dataset Split and Preprocessing: In our experimental setup, we adhere to the dataset configurations used in the GANF study [9]. MTGFlow [39] and MTGFlow_Cluster [40]. Specifically, for the SWaT dataset, we partition the original testing data into 60% for training, 20% for validation, and 20% for testing. For the other datasets, the training partition comprises 60% of the data, while the test partition contains the remaining 40%. The training data are used to train the model, while the validation data are used to tune hyperparameters and the test data are used to evaluate the model's performance. The datasets are preprocessed to remove missing values and normalize the data to a range of [0, 1]. The data are then divided into fixed-length sequences, with a window size of 60

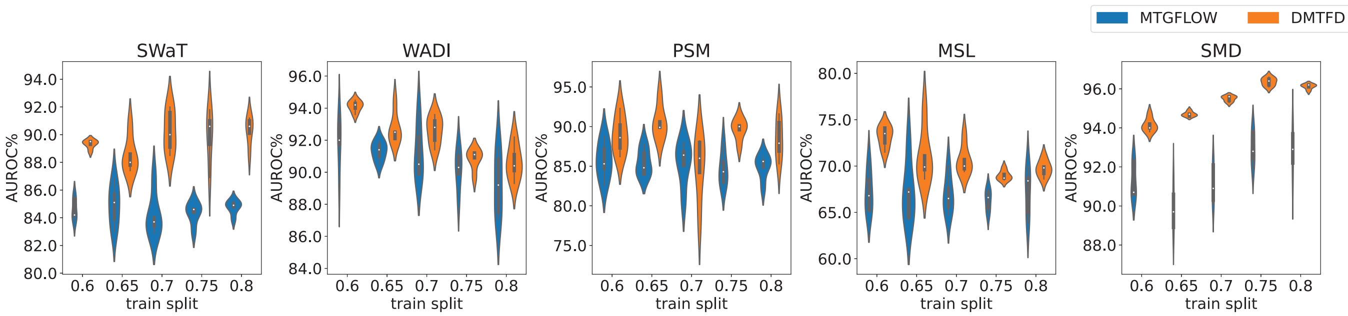

Fig. 4. Violin plot of AUROC scores on different training splits. The x-axis represents the training split ratio, and the y-axis represents the AUROC scores. The DMTFD method consistently outperforms the baseline methods (MTGFLOW) across different training splits, demonstrating its robustness and effectiveness in anomaly detection tasks.

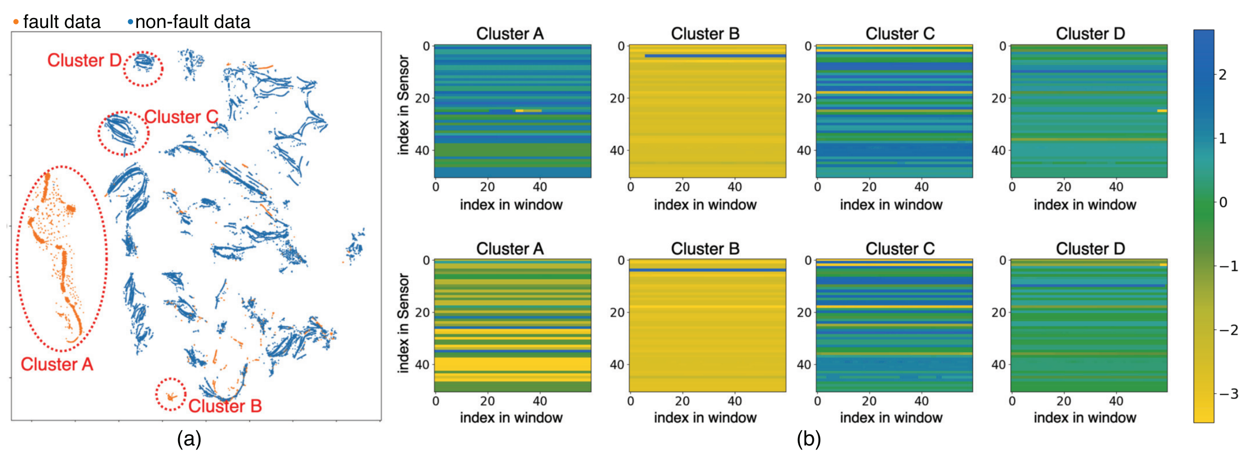

Fig. 5. Visualization of multisubclass data distributions. (a) t-SNE visualization of embedding of DMTFD on SWAT datasets. The color shows the different states of the data. Multiple clusters can be seen in the representation, which is consistent with the multiple manifold assumption of this article. The normal and abnormal states are not a single cluster but can be divided into separate subclusters. (b) Heatmaps of individual data point, each row in the heatmap is the index number of the time, and each column is the index number of the sensor. We selected four subclusters, and two samples from each cluster were randomly selected for presentation. The similarity of samples within subclusters and the difference of samples between subclusters are illustrated.

and a stride of 10. The window size determines the number of time steps in each sequence, while the stride determines the step size between each sequence. The data are then fed into the model for training and evaluation.

2) Implementation Details: For all datasets, we set the window size to 60–80 and the stride size to 10–30. All experiments were run for 400–600 epochs and executed using PyTorch 2.2.1 on an NVIDIA RTX 3090 24GB GPU. Additional specific parameters can be found in the Supplementary Material.

B. Evaluation Metric and Baselines Methods

Following prior work, DMTFD performs window-level anomaly detection, where a window is labeled anomalous if it contains any anomalous point. We evaluate performance using two metrics: area under the receiver operating characteristic (AUROC), which measures overall discriminative ability across thresholds, and **AUPRC**, which is more informative under class imbalance, capturing the tradeoff between precision and recall.

1) Baselines: We compare DMTFD with several state-of-the-art (SOTA) anomaly detection methods. *DROCC* [12] is a robust OCC method assuming locally linear manifolds to avoid representation collapse. *DeepSAD* [26] is a semi-supervised method that detects anomalies via entropy differences in latent distributions. *USAD* [4] employs adversarially trained autoencoders for unsupervised time series anomaly detection. *GANF* [9] enhances normalizing flows using Bayesian networks to model interseries dependencies. *MTGFlow* [39] and *MTGFlow Cluster* [40] utilize dynamic graph structure learning and entity-aware flows for fine-grained density estimation, with the latter introducing clustering for improved accuracy.

C. Fault Detection Performance on Five Benchmark Dataset

First, we discuss the performance advantages of the DMTFD method by comparing results on five common benchmarks. We list the results for the AUROC and AUPRC metrics in Tables I and II, where the numbers in parentheses represent the variance from five different seed experiments. In Tables I and II, the results of DROCC, DeepSAD, USAD, DAGMM, GANF, and MTGFlow are from the article [39].[1] The results of MTGFlow Cluster are from [40].[2] We report on two variants of DMTFD: one (DMTFD) uses the same hyperparameters across all experiments and the other (DMTFD*) involves

[1]https://github.com/zqhang/MTGFLOW

[2]https://github.com/zqhang/MTGFLOW_Cluster

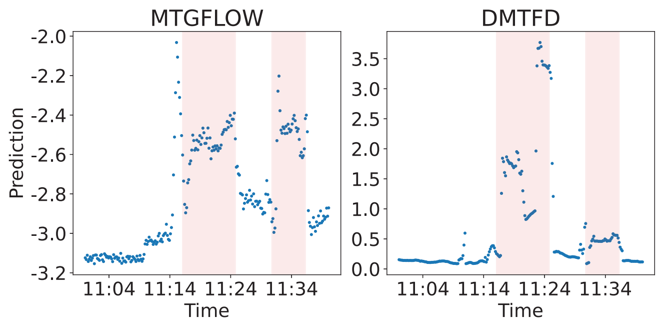

Fig. 6. Point plot of the anomalies predicted outputs of MTGFLOW and proposed DMTFD. The x-axis represents the anomaly index, and the y-axis represents the log likelihood of the anomaly. Anomalous ground truths are marked by a red background. This indicates that DMTFD is able to detect anomalies more accurately and in a timely manner.

hyperparameter tuning specific to each target dataset. The best results are highlighted in bold. To visualize the benefits of DMTFD performance, the ROC and PR curves on four datasets are shown in Fig. 3. In the ROC and PR curves, the DMTFD method consistently outperforms the baseline methods, demonstrating its robustness and effectiveness in anomaly detection tasks.

1) Performance Improvement: In terms of performance improvements, as shown in Tables I and II, the DMTFD method shows significant enhancements compared to SOTA methods. We compare the performance of different methods using a consistent window size (DMTFD) with our optimal results obtained through varying window sizes (DMTFD*) to showcase the maximum potential. The best results are shown in bold. These improvements are evident across both PR and AUC curves, particularly on the SWAT dataset with a 5.4% increase in AUROC and a 7.7% increase in AUPRC, and on the MSL dataset with a 6% increase in AUROC and a 7.0% increase in AUPRC. On average, there is a 4.1% increase in AUROC performance and a 7.0% increase in AUPRC performance. The consistent improvements in both AUROC and AUPRC indicate that the DMTFD method has enhanced performance across multiple aspects.

2) Stability Improvement: As shown in Tables I and II and Fig. 3, the stability of the DMTFD method is also noteworthy, exhibiting lower variance across all test datasets compared to traditional methods. This indicates that the DMTFD method is more robust and less sensitive to random initialization, making it a reliable choice for anomaly detection tasks.

3) Performance Potential: Furthermore, by comparing the results of DMTFD and DMTFD*, we find that conducting hyperparameter searches for each dataset can significantly boost the model performance. This suggests that our method has good hyperparameter adaptability and considerable potential for further performance enhancements.

4) Additional Results on Difference Train/Validation Splits: To further investigate the influence of anomaly contamination rates, we vary training splits to adjust anomalous contamination rates. For all the abovementioned datasets, the training split increases from 60% to 80% with 5% stride. We present an average result over five runs in Fig. 4. Although the anomaly contamination ratio of training dataset rises, the anomaly detection performance of MTGFlow remains at a stable high level. This indicates that the proposed DMTFD method is robust to the anomaly contamination ratio of the training dataset.

TABLE III

Ablation Study. AUROC Comparison on Four Datasets (SWaT, WADI, PSM, and MSL). The Best Results Are Given in Bold. The Results Are Averaged Over Five Runs, and the Standard Deviation Is Shown in Parentheses

Method	SWaT	WADI	PSM	MSL
DMTFD	**90.5(±0.9)**	**94.3(±0.4)**	**89.2(±2.4)**	**75.0(±2.0)**
w. Gaussian	88.9(±1.1)	93.9(±1.7)	86.7(±2.5)	72.1(±0.7)
w/o. MML	84.3(±1.5)	91.7(±1.2)	85.5(±1.3)	70.7(±2.4)
w/o. CL	84.7(±1.3)	90.8(±1.4)	85.9(±1.6)	69.2(±2.1)
w/o. Aug (MTGFlow)	84.8(±1.5)	91.9(±1.1)	85.7(±1.5)	67.2(±1.7)

D. Visualization of Multisubclass Data Distributions

To further investigate the effectiveness of the DMTFD method, we visualize the multisubclass data distributions on the SWAT dataset. As shown in Fig. 5, the t-SNE visualization of the embedding of DMTFD on SWAT datasets reveals multiple clusters, consistent with the multiple manifold assumption of this article. The normal and abnormal states are not a single cluster but can be divided into separate subclusters. The heatmaps of individual data points further illustrate the similarity of samples within subclusters and the differences between samples in different subclusters. This visualization demonstrates the ability of the DMTFD method to capture the diverse patterns present in both normal and abnormal states, enhancing its anomaly detection capabilities.

In addition, we present the point plot of the anomalies predicted outputs of MTGFLOW and the proposed DMTFD in Fig. 6. The x-axis represents the anomaly index and the y-axis represents the log likelihood of the anomaly. Anomalous ground truths are marked by a red background. This indicates that DMTFD is able to detect anomalies more accurately and in a timely manner, outperforming MTGFLOW in terms of anomaly detection performance.

Fig. 7 shows the bar plot of the frequency of normalized anomaly scores on baseline methods and DMTFD. The x-axis represents the anomaly scores and the y-axis represents the frequency of each score. The normalized anomaly scores of DMTFD are significantly lower than those of GANF and MTGFlow, indicating that DMTFD is more effective at distinguishing between normal and abnormal states.

E. Ablation Study: The Effect of SoftCL Loss

To assess the effectiveness of each component designed in our model, we conducted a series of ablation experiments. The results of these experiments are presented in Table III.

We performed controlled experiments to verify the necessity of the SoftCL Loss in the UNDA settings across all four

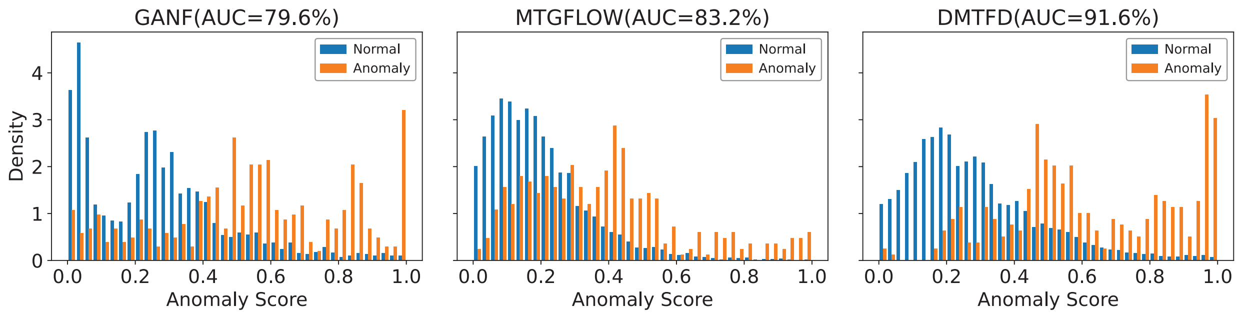

Fig. 7. Bar plot of the frequency of normalized anomaly scores on baseline methods and DMTFD. The x-axis represents the anomaly scores and the y-axis represents the frequency of each score. The normalized anomaly scores of DMTFD are significantly lower than those of GANF and MTGFlow, indicating that DMTFD is more effective at distinguishing between normal and abnormal states.

TABLE IV

PARAMETERS ANALYSIS: LEARNING RATES. TABLE OF AUC SCORES (AUROC%) FOR DIFFERENT LEARNING RATES ON THE SWAT DATASET, INDICATING DMTFD'S SENSITIVITY TO THE LEARNING RATE

lr	0.001	0.005	0.01	0.05	0.1	0.5
AUROC	84.5(±4.2)	**90.9(±0.9)**	90.2(±0.9)	82.3(±1.5)	82.4(±2.7)	79.2(±4.1)

TABLE V

PARAMETERS ANALYSIS: NUMBER OF EPOCHS. TABLE OF AUC SCORES (AUROC%) FOR DIFFERENT EPOCHS ON THE SWAT DATASET, INDICATING DMTFD'S SENSITIVITY TO THE NUMBER OF EPOCHS

epoch	40	100	200	400	500	1000
AUROC	81.4(±2.9)	85.8(±2.9)	89.8(±1.4)	**90.2(±0.9)**	**90.2(±0.9)**	**90.2(±0.9)**

TABLE VI

PARAMETERS ANALYSIS: HYPERPARAMETER ν. TABLE OF (AUROC%) SCORES (AUROC%) FOR DIFFERENT ν ON THE SWAT DATASET, INDICATING DMTFD'S SENSITIVITY TO THE HYPERPARAMETER ν

ν	0.005	0.01	0.05	0.1
AUROC	88.4(±1.8)	90.2(±0.9)	90.5(±1.3)	**90.6(±2.6)**

datasets. The performance of the proposed DMTFD method is denoted as *DMTFD* in Table A-5 in the Supplementary Material. The variant *w/o. SoftCL* represents the model performance with the SoftCL loss component, $\mathcal{L}_{\text{MML}}(\mathbf{x}_i^t, y_i{}^t)$, removed from the overall loss function of DMTFD. The variant *w. CL* indicates the model performance when the SoftCL loss is replaced by a typical contrastive loss, $\mathcal{L}_{\text{CCL}}$, as defined in (6). The results clearly demonstrate that the SoftCL Loss significantly outperforms the traditional CL loss. We attribute the inferior performance of $\mathcal{L}_{\text{CCL}}$ to its inability to adequately address the view-noise caused by domain bias.

F. Hyperparameter Robustness

1) Hyperparameter Robustness: Window Size and Number of Blocks: Table A-5 in the Supplementary Material presents an ablation study investigating the robustness of hyperparameters, focusing on the window size and number of blocks. Three distinct datasets, SWaT, WADI, and PSM, along with MSL, are evaluated using DMTFD. Our method showcases promising results, with consistently competitive AUROC scores across varying hyperparameter configurations. Notably, the standard deviations accompanying AUROC values indicate a high degree of stability, underscoring the reliability of our approach. Analysis of the table reveals that optimal configurations often coincide with larger window sizes and moderate block numbers, suggesting a preference for capturing broader temporal contexts while maintaining computational efficiency. Additionally, trends indicate that as the number of blocks increases, there is a discernible improvement in performance, albeit with diminishing returns beyond a certain threshold. This observation highlights the importance of carefully balancing model complexity with computational resources. Overall, the DMTFD method demonstrates robustness and effectiveness in anomaly detection tasks.

2) Hyperparameter Robustness: Learning Rate, ν, and Number of Epoch: In order to further investigate the effectiveness of MTGFlow, we give a detailed analysis based on the SWaT dataset. We conduct ablation studies on the learning rate, ν, and the number of epochs. The results are shown in Tables IV–VII. The results show that the performance of MTGFlow is relatively stable across different learning rates, ν, and the number of epochs. This indicates that MTGFlow is robust to hyperparameter changes and can achieve good performance with a wide range of hyperparameters.

TABLE VII

PARAMETERS ANALYSIS: WEIGHT OF NEGATIVE (NE) SAMPLE AND POSITIVE (PO) SAMPLE. TABLE OF AUC SCORES (AUROC%) FOR DIFFERENT TOPOLOGICAL LOSS RATIOS ON THE SWAT DATASET

ne \ po	0.5	1	2	3	4	5
1	91.2(±1.0)	90.2(±3.1)	91.3(±1.7)	89.8(±1.8)	90.9(±1.1)	89.8(±1.8)
2	89.0(±2.0)	90.0(±2.3)	90.1(±0.7)	90.6(±1.6)	90.6(±2.3)	91.2(±1.9)
3	89.8(±2.8)	90.3(±2.5)	91.2(±1.7)	88.6(±2.8)	91.3(±0.9)	90.1(±1.8)
4	89.9(±1.1)	89.1(±2.6)	90.3(±1.0)	88.7(±2.5)	90.1(±0.8)	90.4(±2.7)
5	89.2(±3.2)	90.2(±0.9)	92.1(±1.1)	89.0(±2.2)	90.2(±0.8)	90.3(±1.1)
6	89.3(±0.9)	90.9(±1.4)	89.4(±1.2)	90.0(±1.3)	89.5(±2.2)	90.4(±1.9)

VI. CONCLUSION

In this work, we propose DMTFD, an unsupervised framework for fault detection in multivariate time series. To overcome the limitations of Gaussian assumptions, DMTFD models data with multi-Gaussian representations. A neighbor-based augmentation strategy is designed to generate positive pairs for learning, while a representation head enables optimization based on local streamform similarities. This enhances the separation between normal and anomalous states and broadens anomaly coverage. Experiments on standard benchmarks (e.g., SWaT and WADI) demonstrate that DMTFD achieves superior AUC and PR scores, with lower false alarm rates and improved detection accuracy. Visualizations also validate the effectiveness of the multi-Gaussian modeling. Although our current focus is unsupervised fault detection, DMTFD can be extended to remaining useful life (RUL) prediction. Its pretrained embeddings provide strong priors for temporal degradation modeling and, with minimal finetuning, can support few-shot RUL forecasting. We leave multitask extensions integrating fault detection and RUL estimation for future work.

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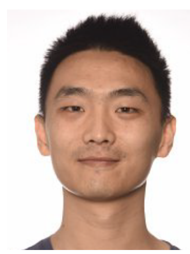

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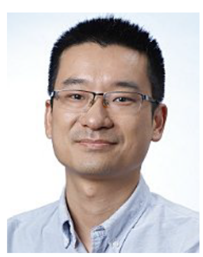

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